Question 2: Testing

Assignment

> Marktplaats has decided on implementing a product feature that we think will drive the number of Car ads with leads: How would you ideally set up the test whether this initiative has improved this metric?
>
> - Why do you choose this method?
> - What are the risks / challenges?
> - Use the dataset attached to examine the test groups:
>   - What is the result of your analysis?
>   - What insights did you gain?

1. Business Question and Test Design

Primary KPI

Share of Car Ads With at Least One Lead.

`has_any_lead = telclicks > 0 OR bids > 0 OR n_asq > 0 OR webclicks > 0`

Test Design

Use a randomized A/B test to answer whether the new product feature increases the share of car ads that receive at least one lead.

Why this method:

- Random assignment is the cleanest way to estimate whether the feature caused a change in lead rate.
- It makes treatment and control ads more comparable before exposure.
- It is better than a before/after comparison, which can be affected by seasonality, buyer demand, or changes in ad supply.

Preferred randomization level: ad-level, assuming the feature affects individual ads and does not strongly change seller behavior across ads. If the feature is seller-side, seller-level randomization may be safer.

Treatment Assumption

- I assume B is the treatment group and A is control.
- If the assignment were reversed, the product interpretation would reverse too.

Design Risks

Before treating the A/B gap as causal, I would validate four areas:

- Assignment integrity: confirm A/B assignment was random, consistently applied, and close to the planned allocation. Any assignment bug or sample-ratio mismatch could bias the result.
- Group comparability: check whether A and B are balanced on key ad and car attributes such as price, mileage, age, seller type, and placement timing. The dataset shows B differs from A on some dimensions, so part of the lift could reflect mix rather than product impact.
- Measurement consistency: confirm lead events are logged the same way in both groups and that missing, duplicate, or unassigned rows are handled using rules defined before reading outcomes.
- Marketplace side effects: monitor whether the feature shifts leads away from control ads or increases low-quality leads, duplicate leads, complaints, or seller/buyer friction.

In [48]:

```
from pathlib import Path
import sys
import importlib

import pandas as pd
from IPython.display import display

helper_dir = Path.cwd()
if not (helper_dir / "functions.py").exists():
    helper_dir = Path.cwd() / "assignments" / "data_analyst"
if str(helper_dir) not in sys.path:
    sys.path.insert(0, str(helper_dir))

import functions as helpers

helpers = importlib.reload(helpers)

pd.set_option("display.float_format", "{:,.3f}".format)
pd.set_option("display.max_columns", 80)
helpers.plt.style.use("seaborn-v0_8-whitegrid")
```

2. Data Readiness

Before comparing outcomes, I check whether the lead metrics and ad identifiers are reliable enough for analysis:

- Lead metrics are clean enough for analysis: almost no missing values and no negative values.
- Missing lead metrics are minimal: 9 rows have missing values in at least one lead column, so treating missing lead counts as zero has negligible impact.
- Duplicate ad IDs are negligible: 10 rows, checked later as a sensitivity test.

In [50]:
```
raw_data = helpers.read_ab_test_data(helpers.DATA_PATH)
ab_data, excluded_data = helpers.prepare_ab_data(raw_data)
segmented_data = helpers.add_segments(ab_data)

print(f"Rows: {len(raw_data):,}")
```

```
Rows: 183,062
```

Dataset Shape and Identifier Quality

Quick check of row count, column count, full duplicates, and repeated ad IDs.

In [51]:
```
display(helpers.style_table(helpers.data_quality_checks(raw_data)))
```

	Check	Value
0	Rows	183,062
1	Columns	22
2	Fully Duplicated Rows	0
3	Rows With Duplicated Ad ID	10
4	Unique Ad IDs	183,057

Undocumented Fields

Columns outside the core assignment schema are documented before deciding whether to use them.

In [52]:
```
display(helpers.style_table(helpers.undocumented_column_notes(raw_data)))
```

	Column	Note
0	Color	used as an optional car-color dimension
1	Mileage	used as an optional mileage dimension
2	Days Live	used as an optional ad-tenure dimension
3	L2	not used; meaning is unclear

Lead Metric Quality

Checks whether lead-event columns have missing, negative, or extreme values.

In [53]:
```
display(helpers.style_table(helpers.lead_metric_quality(raw_data)))
```

	Metric	Missing Rows	Missing %	Negative Rows	Zero Rows	Max Value
0	Tel Clicks	9	0.000	0	125,624	162
1	Bids	9	0.000	0	152,145	124
2	Questions	0	0.000	0	122,577	282
3	Web Clicks	9	0.000	0	85,122	410

Missing Values

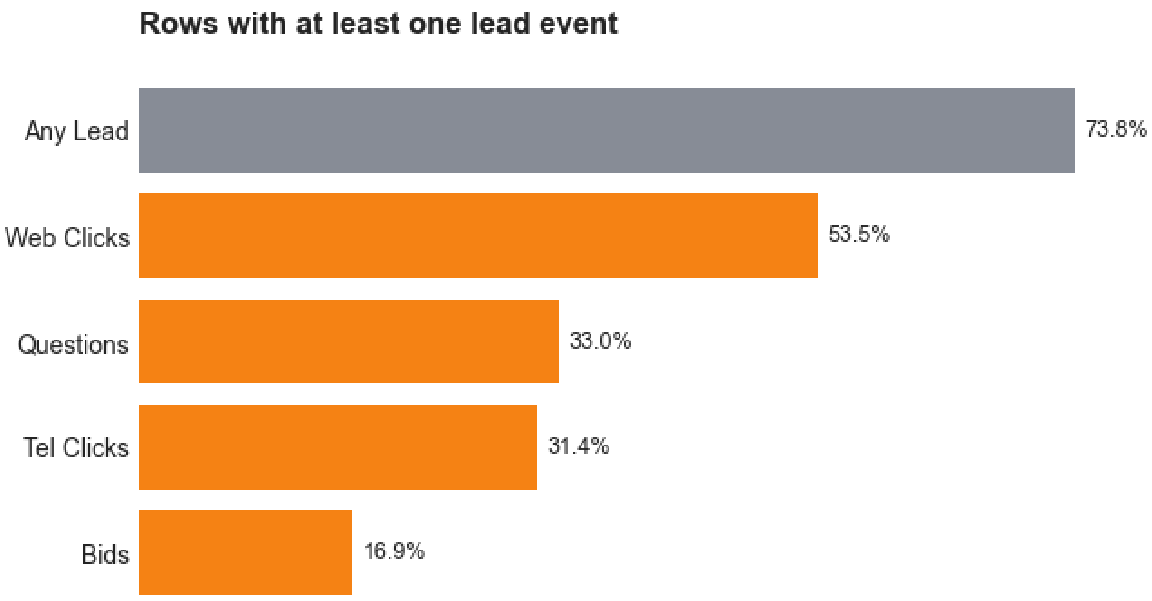
Shows which optional columns have missing values.

In [54]:
```
display(helpers.style_table(helpers.column_quality_summary(raw_data).quer
```

	Column	Dtype	Missing Rows	Missing %	Unique
0	Energy Label	str	49,384	0.270	7
1	Body Type	str	17,851	0.098	8
2	Seats	Int64	11,743	0.064	16
4	Doors	Int64	5,375	0.029	9
5	Model	str	925	0.005	2,447
6	Emissions	Int64	191	0.001	356
7	Bids	Int64	9	0.000	42
8	Tel Clicks	Int64	9	0.000	65
9	Web Clicks	Int64	9	0.000	154
10	Price	Int64	7	0.000	7,522

Next, check lead-event coverage. The chart compares each channel with the combined `any lead` KPI.

In [55]:
```
helpers.plot_lead_channel_coverage(raw_data)
helpers.plt.show()
```



- The chart shows each lead channel is much smaller than `Any Lead`.
- This supports using the combined KPI as the primary metric.
- Later checks still show lead volume by channel.

- Lead metrics have very few missing values and no negative values.
- Some undocumented fields are useful for balance and segment checks.
- `l2` is excluded because its meaning is unclear.

Duplicated Ad IDs

There are 5 duplicated `src_ad_id` values, covering 10 rows.

Duplicated Ad Rows

Lists the small set of repeated ad IDs for direct inspection.

In [56]:
```
display(helpers.style_table(helpers.duplicated_ad_id_rows(raw_data)))
```

	Ad ID	Tel Clicks	Bids	Body Type	Photos	Doors	Questions	Build Year	Emissic
0	1011548826	0	0	MPV	18	5	0	2003	1
1	1011548826	0	0	Hatchback (3/5-deurs)	14	4	0	2005	1
2	1027581466	0	0	Hatchback (3/5-deurs)	13	5	0	2010	
3	1027581466	0	0	MPV	24	4	0	2008	1
4	1045227572	0	0	Hatchback (3/5-deurs)	24	5	0	2012	1
5	1045227572	0	0	Terreinwagen	24	5	0	2013	1
6	1054003502	0	0	Stationwagon	24	5	0	2005	2
7	1054003502	1	0	Sedan (2/4-deurs)	11	4	0	2006	1
8	1072325510	0	0	Hatchback (3/5-deurs)	12	3	0	2013	
9	1072325510	0	0	Cabriolet	24	2	0	2000	2

Duplicate-ID Differences

Summarizes which fields differ within each repeated ad ID.

In [57]:
```
display(helpers.style_table(helpers.duplicated_ad_id_differences(raw_data
```

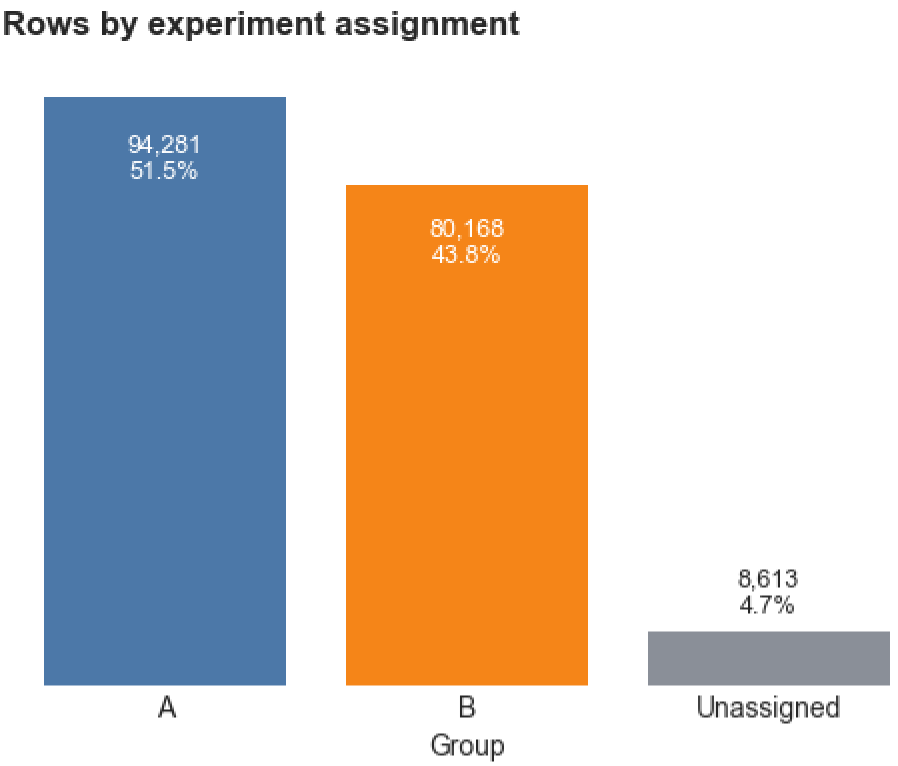
	Ad ID	Rows	Groups Seen	Varying Columns
0	1011548826	2	A, B	carrosserie, photo_cnt, aantaldeuren, bouwjaar, emissie, energielabel, brand, vermogen, webclicks, model, price, group
1	1027581466	2	A, B	carrosserie, photo_cnt, aantaldeuren, bouwjaar, emissie, energielabel, brand, vermogen, webclicks, model, price, group
2	1045227572	2	A, B	carrosserie, bouwjaar, emissie, energielabel, brand, vermogen, webclicks, model, price, group
3	1054003502	2	A	telclicks, carrosserie, photo_cnt, aantaldeuren, bouwjaar, emissie, energielabel, brand, vermogen, model, price
4	1072325510	2	A, B	carrosserie, photo_cnt, aantaldeuren, bouwjaar, emissie, energielabel, brand, vermogen, webclicks, model, price, group

The paired duplicate-ID rows differ on car attributes. Decimal-looking ad IDs are preserved as text, and these duplicates are exact repeated IDs rather than a truncation issue. I keep them in the primary analysis because they are only 10 rows, then test sensitivity later by excluding them.

3. Group Assignment Check

- Raw groups: A, B, and Unassigned.
- 95.3% of rows are assigned to A or B; 4.7% are unassigned and excluded from A-vs-B outcome analysis.
- If the intended design was 50/50, the A/B split should be checked for sample-ratio mismatch before causal interpretation.

In [58]:
```
helpers.plot_group_sizes(raw_data)
helpers.plt.show()
```



The dataset has 174,449 valid A/B rows and 8,613 unassigned rows. If the intended split was 50/50, the A/B allocation should be checked with the experiment owner before interpreting the result causally.

4. Balance Check

- B differs from A on key numeric car/ad dimensions, especially price, car age, and mileage.
- Numeric balance is reported with both percent differences and standardized differences.

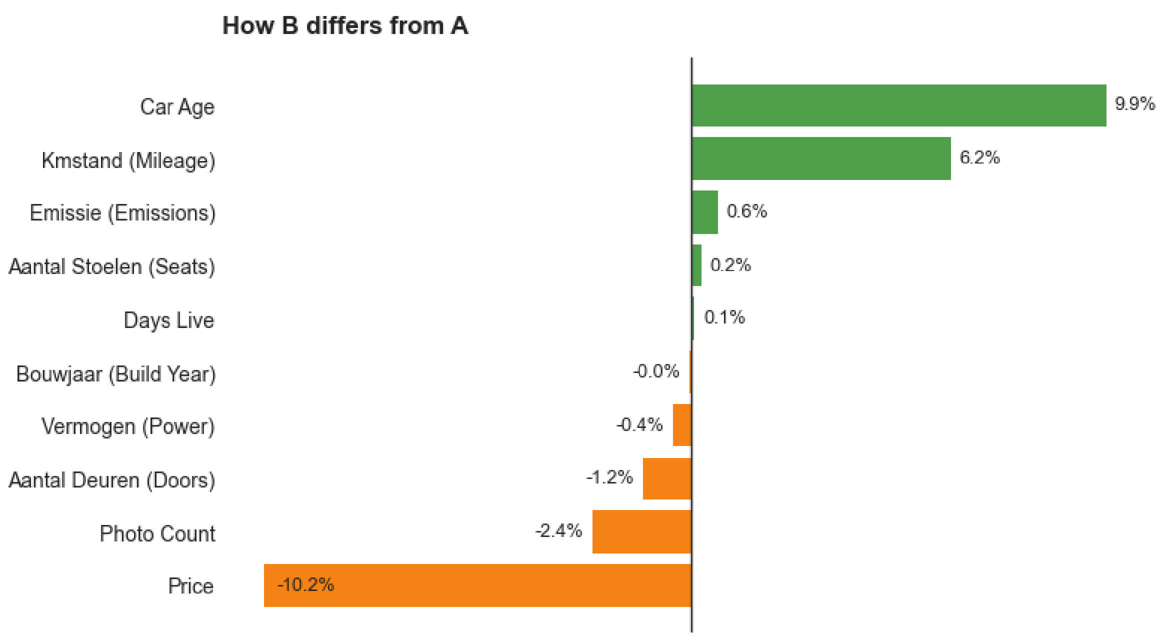
- I also check top categorical dimensions, because group mix can affect lead rates even when the primary KPI difference is statistically significant.

In [59]:
```
display(helpers.style_table(helpers.numeric_balance_summary(ab_data)))
display(helpers.style_table(helpers.categorical_balance_overview(ab_data)
```

group	Dimension	A	B	Diff B-A	% Diff	Std Diff
0	Price	106,457.104	95,563.256	-10,893.849	-0.102	-0.012
1	Mileage	140,764.544	149,516.859	8,752.315	0.062	0.094
2	Days Live	27.624	27.646	0.022	0.001	0.001
3	Photos	15.742	15.366	-0.376	-0.024	-0.058
4	Build Year	2007.560	2006.721	-0.839	-0.000	-0.155
5	Doors	4.270	4.220	-0.049	-0.012	-0.050
6	Seats	4.735	4.747	0.012	0.002	0.013
7	Emissions	129.123	129.950	0.827	0.006	0.012
8	Power	85.787	85.420	-0.367	-0.004	-0.009
9	Car Age	8.440	9.279	0.839	0.099	0.155

	Dimension	Largest Category Gap	Share A	Share B	Share Diff
0	Brand	FORD	0.074	0.071	-0.003
1	Model	Corsa	0.016	0.017	0.001
2	Color	Wit	0.110	0.100	-0.010
3	Body Type	Terreinwagen	0.058	0.053	-0.005
4	Energy Label	Missing	0.257	0.285	0.028

In [60]:
```
helpers.plot_numeric_balance(ab_data)
helpers.plt.show()
```

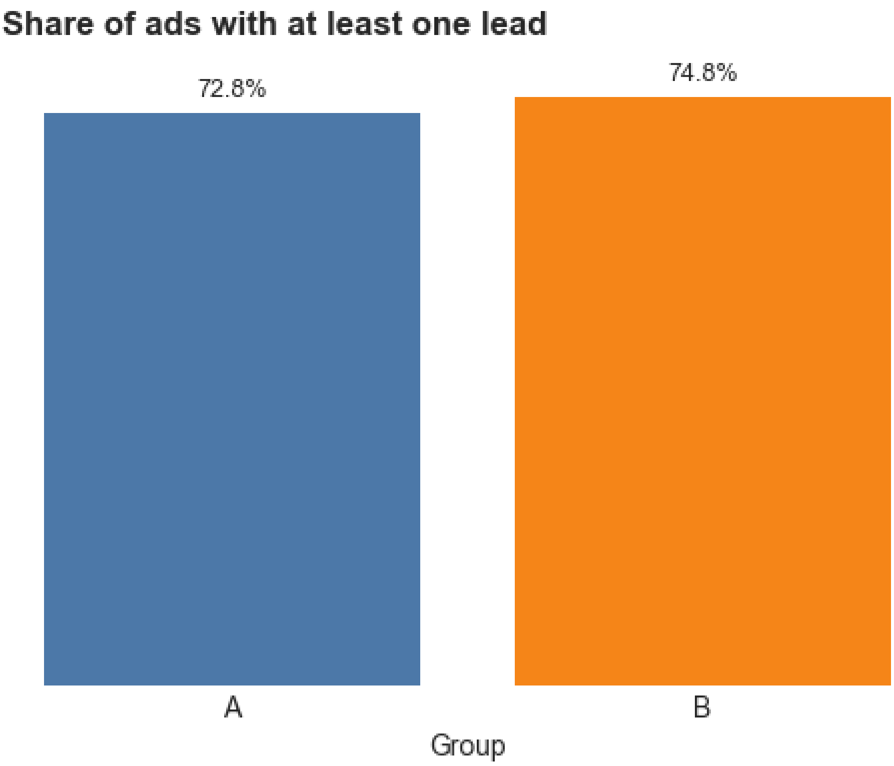


5. Primary Result and Recommendation

- Assuming B is the group that received the feature, B has a 74.8% lead rate vs 72.8% for A.
- Difference: +2.0 percentage points; statistically significant ($p < 0.001$).
- B also has more total leads per ad.
- Caveat: groups are not perfectly balanced, so some of the gap could reflect group mix rather than only the product feature.
- Recommendation: B is directionally positive, but investigate both the assignment imbalance and segment mix imbalance before rollout; also confirm which group received the feature.

In [61]:
```
outcomes = helpers.lead_outcome_summary(ab_data)

helpers.plot_lead_rate(outcomes)
helpers.plt.show()
```



Statistical Inference

- The chart shows the primary KPI directly.
- The table below focuses on lift, confidence interval, and statistical inference.
- I use a **two-sided** two-proportion z-test. Even though the feature is expected to help, a product change could theoretically also hurt leads, so I test for a difference in either direction.
- Given the assignment and balance concerns above, I interpret statistical significance as evidence that the raw A/B gap is unlikely to be sampling noise, not as causal proof by itself.

- Absolute lift: B minus A, in percentage points.
- Relative lift: absolute lift divided by A lead rate.

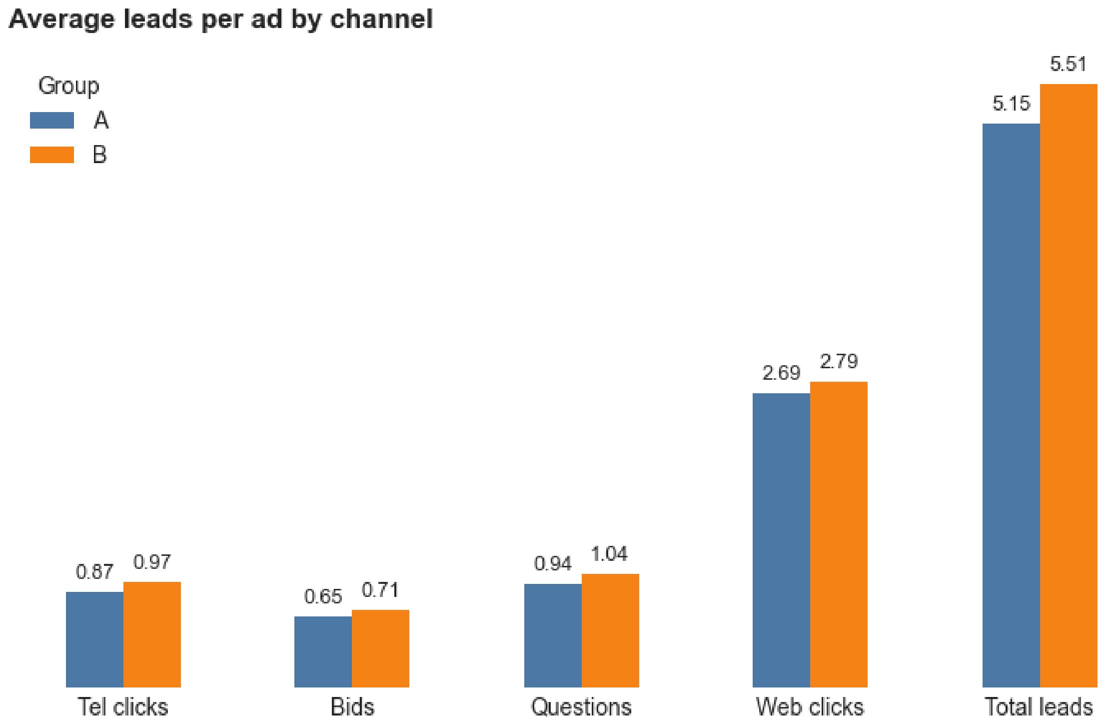
In [62]:
```
display(helpers.lift_summary_table(ab_data).style.hide(axis="index"))
```

Metric	Value
Absolute lead-rate lift	2.0 pp
95% CI for absolute lift	1.6 to 2.4 pp
Relative lift vs A	2.8%
p-value	<0.001
Significance level	5%

6. Lead Volume Checks

- B also has higher average lead volume per ad.
- Channel checks show whether the lift is broad or concentrated in one lead type.

In [63]:
```
helpers.plot_average_leads_by_channel(outcomes)
helpers.plt.show()
```



- B is higher on every lead channel and total leads per ad.
- The lift is spread across channels, not driven by one large channel spike.

7. Robustness Check

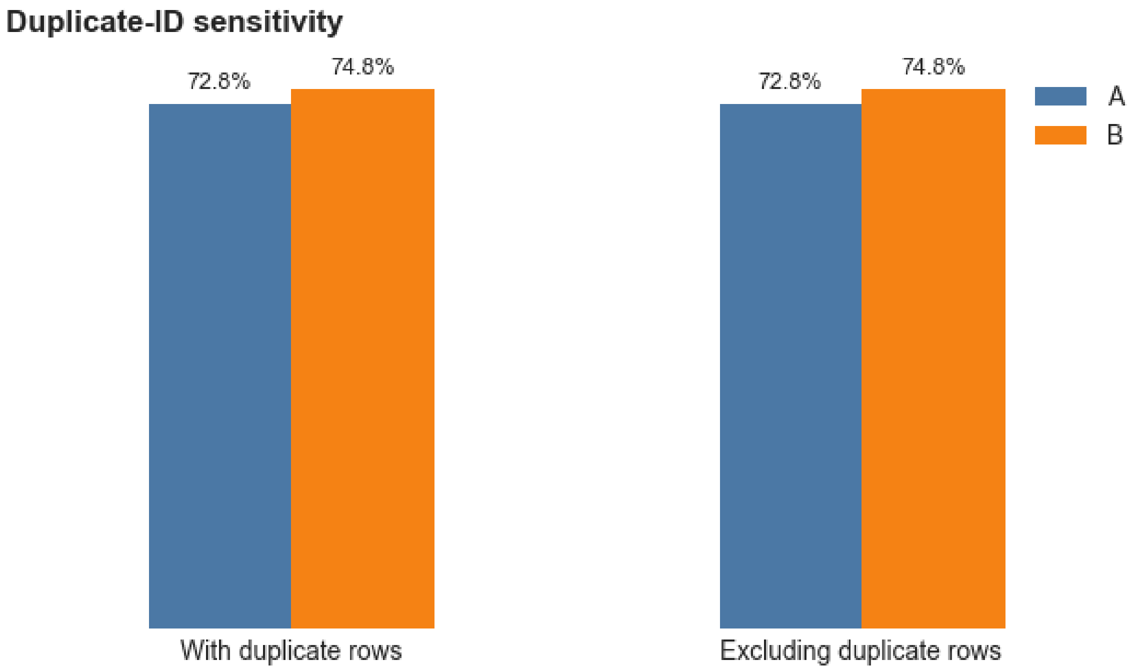
Purpose: verify that the duplicate-ID issue found during data readiness does not drive the primary conclusion.

In [64]:
```
display(helpers.style_table(helpers.duplicate_id_sensitivity(ab_data)))
```

	Metric	With Dup IDs	Excl Dup IDs
0	Rows	174,449	174,439
1	Rows Dropped	0	10
2	% Rows Dropped	0	0.000
3	A Lead Rate	0.728	0.728
4	B Lead Rate	0.748	0.748
5	Abs Lead-Rate Lift	0.020	0.020
6	Relative Lift vs A	0.028	0.028

Excluding duplicated `src_ad_id` rows does not materially change the A/B lead-rate comparison. The result is robust to this specific data-quality issue.

In [65]:
```
helpers.plot_duplicate_id_sensitivity(ab_data)
helpers.plt.show()
```



Removing duplicated ad IDs leaves the A and B lead rates essentially unchanged. The duplicate-ID issue is worth documenting, but it does not drive the main result.